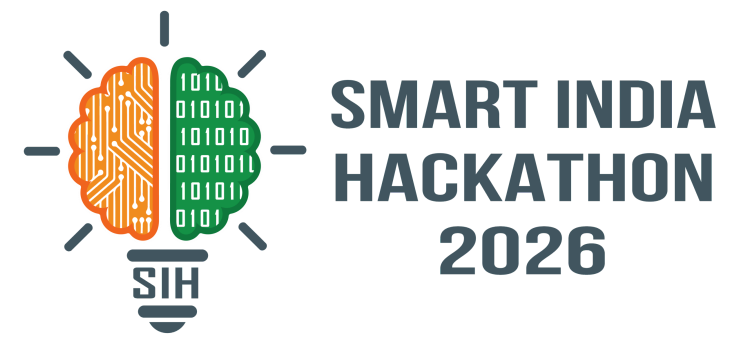




SMART INDIA HACKATHON 2026



- Problem Statement ID - **SIH26027**
- Problem Statement Title - **AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways**
- Theme - **Transportation & Logistics**
- PS Category - **Software**
- Team ID - **137710**
- Team Name - **TRIKAAL**

RAKSHA - BLOCK

(Predict. Optimize. Protect.)

"One shared, risk-ranked, human-approved block plan for Engineering, S&T and TRD."

PROBLEM EXISTING

- **Siloed Planning** : Engineering, S&T and TRD each raise block requests separately through BDMS.
- **Scattered Data** : Defects and overdue work sit in TMS, SMMS and TDMS; corridor availability sits in COA.
- **No Common Priority** : Each department ranks its own work; there is no shared risk measure.
- **Manual Coordination** : Planning is decentralized and manual, so overlaps are found late.

PROPOSED SOLUTION

- **Smart Bundling** : CP-SAT merges compatible Engineering, S&T and TRD tasks (same section, date and line, no shared machinery, overlapping windows) into one shared block.
- **AI Prioritization** : A 0–100 risk score from severity, days overdue, asset age, past failures (LightGBM + SHAP).
- **On-demand re-optimization** : The controller re-runs the optimizer on the current pending requests at any time.
- **Human-in-the-Loop** : AI recommends; Section Controller approves, modifies or rejects, with a reason recorded.

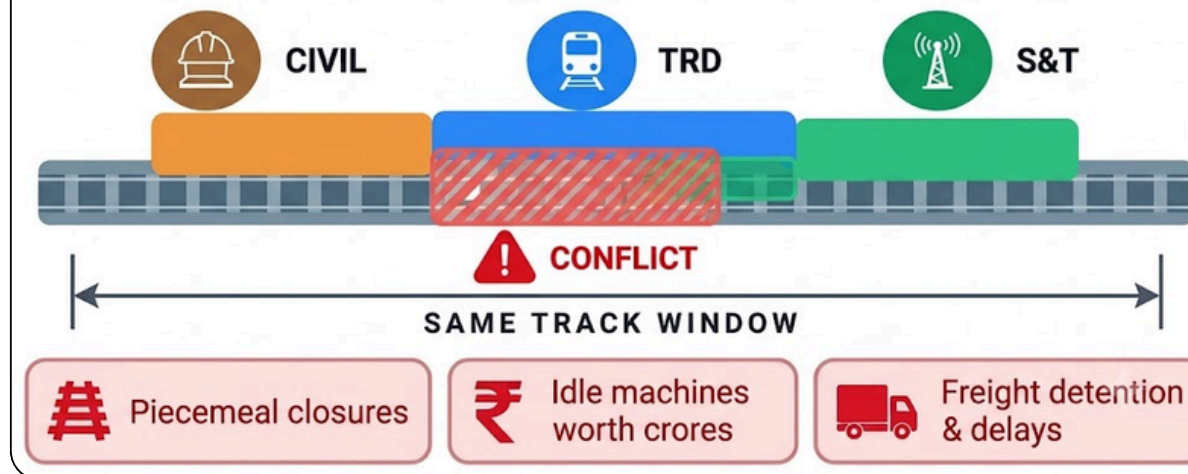
WHAT MAKES RAKSHA-BLOCK DIFFERENT

- **Block-as-Resource** : Block time is planned as one shared resource across three departments.
- **Rule-Validated** : Every bundle is checked against G&SR safety rules - headway, traction isolation.
- **Live sync** : Requests and decisions update on every officer's device without a page refresh. Hu
- **Raksha-Saarthi AI** : Chat answers from the live board (requests, timetable, capacity).
- **Integration-Ready** : Adapters for TMS, SMMS, TDMS and COA with a live source-status panel.

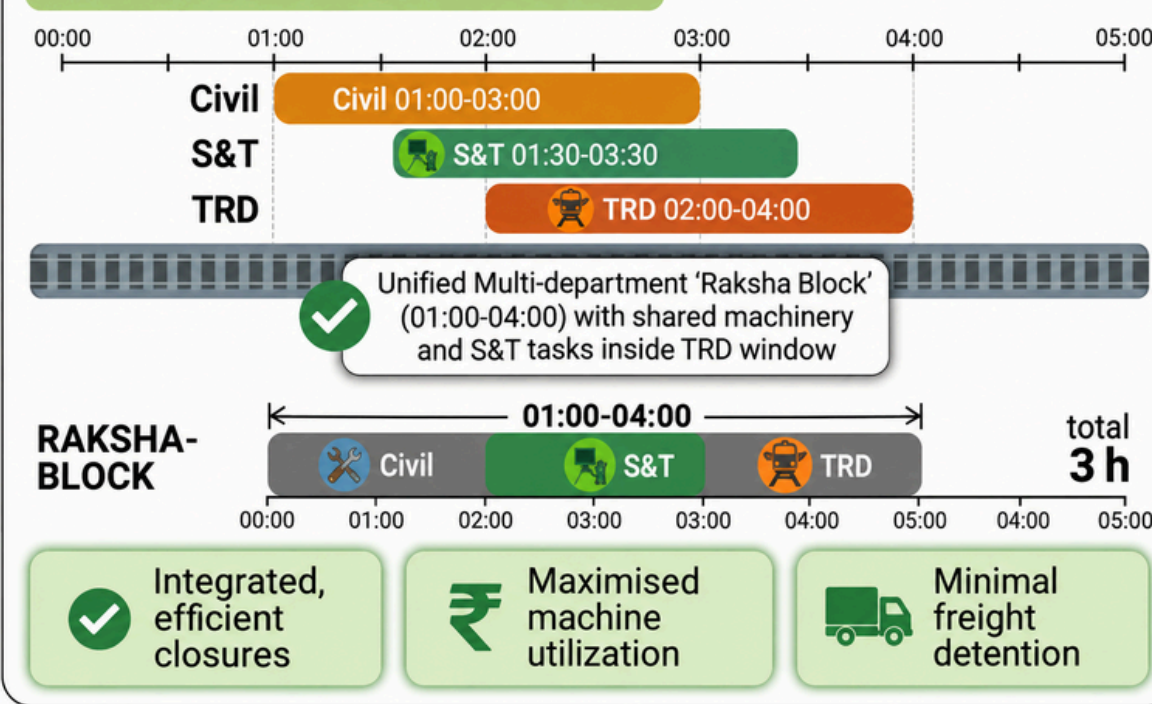


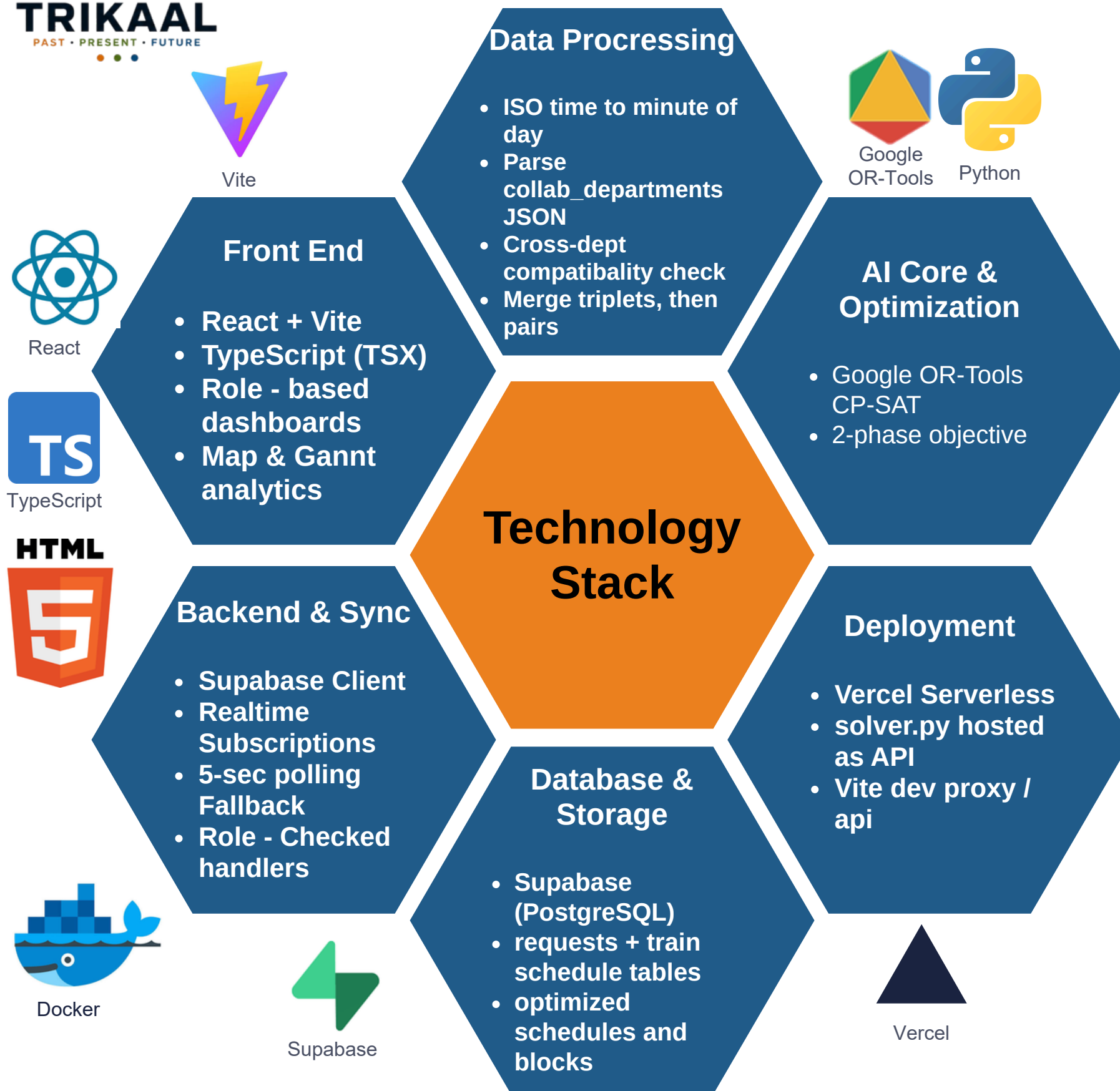
TODAY — THE PROBLEM

Every department books track windows independently through BDMS — nobody sees the full picture.



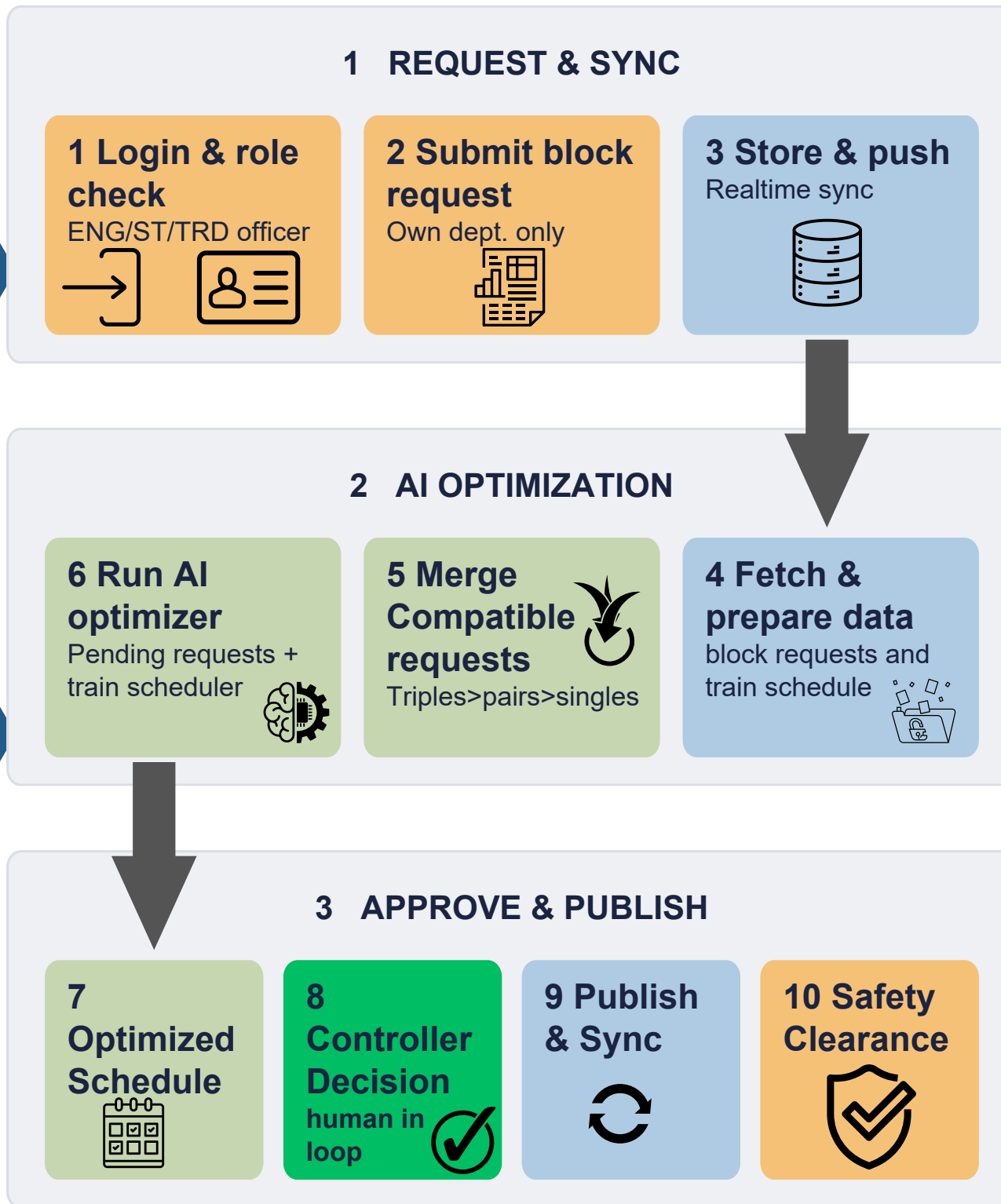
RAKSHA BLOCK — THE SOLUTION





Data Flow of RAKSHA-BLOCK

● Officer ● Controller ● Supabase ● CP-SAT

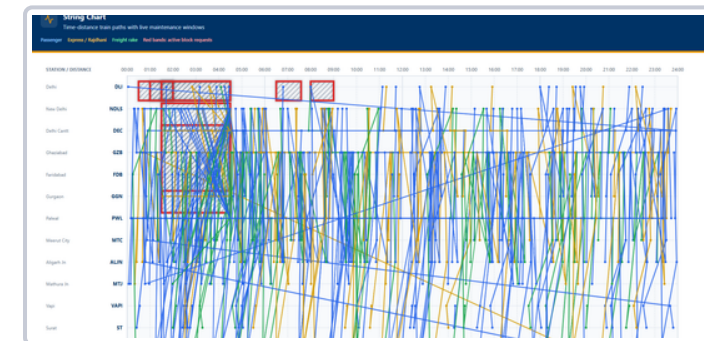


WORKING PROTOTYPE - LIVE ✓

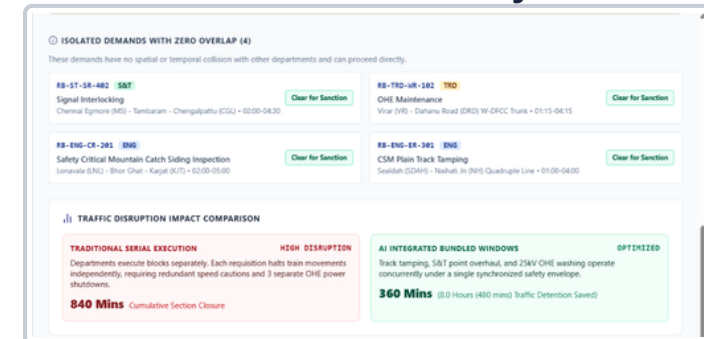


raksha-block-trikaal.vercel.app

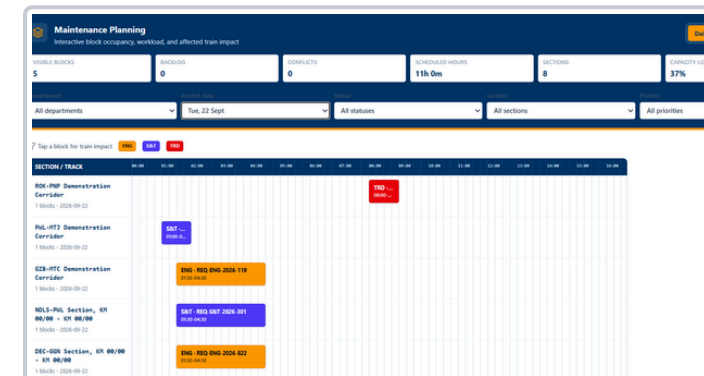
STRING CHART



CP-SAT Solver Analysis



GANTT CHART



FEASIBILITY AND VIABILITY

TECHNICAL

Open-source, CPU-only

- ✓ OR-Tools CP-SAT + LightGBM/SHAP run on CPU.
- ✓ TMS, SMMS, TDMS, COA adapters feed one schema.

OPERATIONAL

Advisory overlay on today's process

- ✓ Departments request as now; Controller approves, modifies or rejects.
- ✓ Role-based portals; no change to field protocols.

ECONOMIC

Software only

- ✓ No trackside hardware or IoT retrofit.
- ✓ Cloud stack-cost per division per year is < **1.5 Lakhs/year** + Zero hardware Cost.

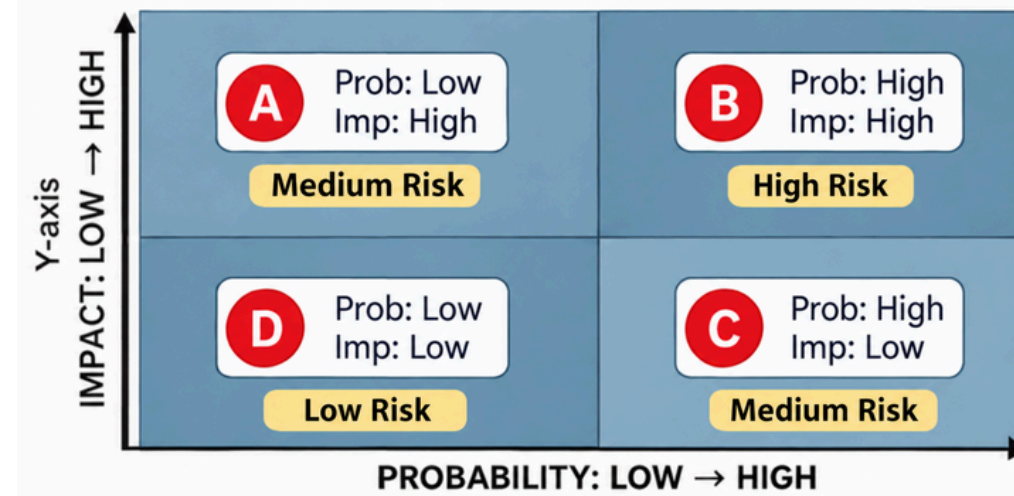
SAFETY

Safety rules stay with the railway

- ✓ Premium-train windows blocked before a bundle is proposed.
- ✓ Human approval with reason; Co-Pilot is read-only; safety checkout closes each block.

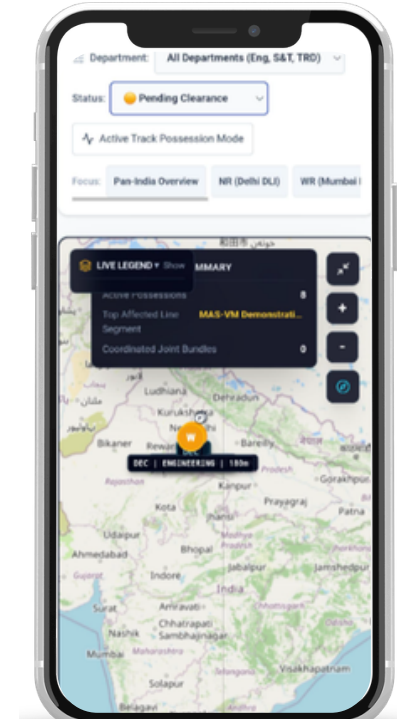


RISK MAP — CHALLENGES & MITIGATIONS



A = Solver Scale
B = Live Systems Access

C = Model Quality
D = Department Resistance



DETAILED BREAKDOWN

! Live systems (TMS, SMMS, TDMS, COA) are internal.

! Solver speed at scale

! Model trained on synthetic outcomes

! Departments resist a new tool

MITIGATION / STRATEGY

➔ **SYNTHETIC DATA PILOT:** Use matching schema synthetic data. Pilot read-only export after Railway approval.

➔ **MEASURED PERFORMANCE:** 7-day plan optimized in < **2 s**. Interval model for real-time bundling.

➔ **MODEL PERFORMANCE & RETRAINING:** LightGBM outperforms baselines (ROC-AUC 0.95). Shadow mode retraining on real failure data.

➔ **ADVISORY OVERLAY & DECISION-MAKING:** Common shared queue and shared risk score. Maintain Section Controller authority.

 | **RAKSHA recommends;** the Section Controller decides. No block is granted by the software.

TARGET AUDIENCE



SOLUTION BENEFITS

SOCIAL

- ✓ Fewer separate blocks means fewer repeated protection set-ups for track and OHE crews (**-18%, illustrative synthetic scenario**).
- ✓ All three departments see one queue and one risk score; every decision has a recorded reason.

ECONOMIC

- ✓ Fewer possession hours (**-12%, illustrative synthetic scenario**) leaves more of each night free for train paths.
- ✓ Software-only overlay: no trackside hardware.

ENVIRONMENTAL

- ✓ Fewer block windows overlapping trains (**-15% touched, illustrative synthetic scenario**) means fewer regulated trains.
- ✓ Requests, approvals and clearances are recorded digitally with PDF/CSV export.

Better Asset Availability



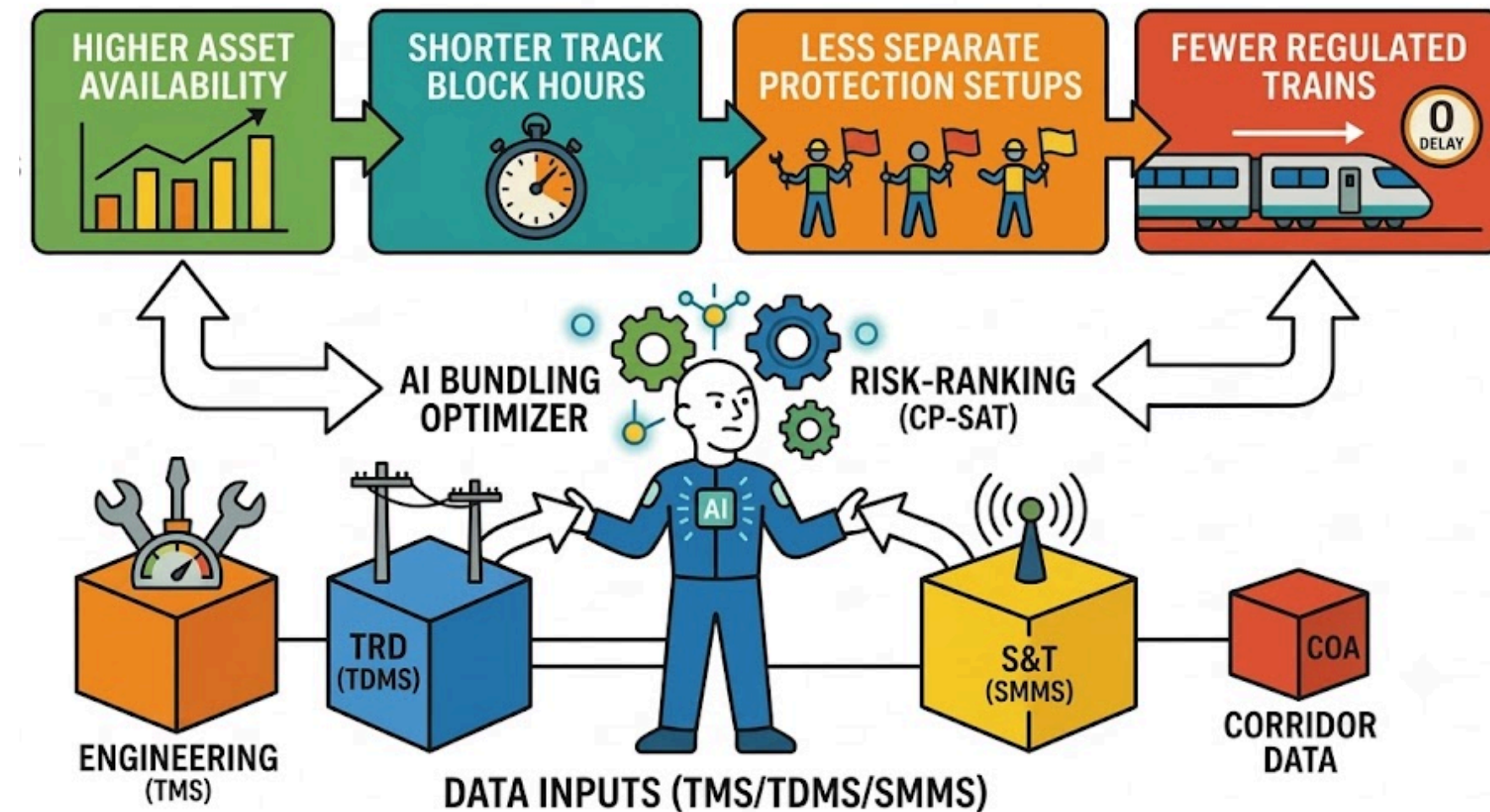
Fewer Protection Setups



Less Train Disruption



RAKSHA-BLOCK: AI MAINTENANCE OPTIMIZATION IMPACT



STANDARDS & SAFETY BASIS



General Rules (1976) + Subsidiary Rules (G&SR)

Block working and track-machine block requisition rules



https://www.nfrlyconstruction.org/uploads/File/Manuals/02042023-13_141_SCR_G_SR.pdf_203600075.pdf



Indian Railways Permanent Way Manual (2020)

P-Way duties and coordination between departments



<https://iricen.gov.in/iricen/CodeManualNew.jsp>



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Ministry of Railways · Software ·
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<https://www.sih.gov.in>

METHODS & TECHNOLOGY



Google OR-Tools CP-SAT

Constraint solver behind day-of bundling and weekly/monthly planning



https://developers.google.com/optimization/cp/cp_solver



Railway maintenance scheduling research

Peralta et al. (2018), J. Comput. Civ.
Eng. 32(3)



[https://ascelibrary.org/doi/10.1061/\(ASCE\)CP.1943-5487.0000757](https://ascelibrary.org/doi/10.1061/(ASCE)CP.1943-5487.0000757)



LightGBM (Ke et al., 2017) · SHAP (Lundberg & Lee, 2017)

Failure-risk model and its explanations



<https://proceedings.neurips.cc/paper/2017/file/6449f44a102fde848669bdd9eb6b76fa-Paper.pdf>

PROJECT RESOURCES



Problem Understanding & Animated Explanation

Short visual explanation of the railway block-planning problem.



<https://youtu.be/xhBWHfrbTVQ?si=yyxli00jKkSJXp-w>



GitHub Repository

Source code, synthetic data, models
& implementation



https://github.com/khushhh-14/Raksha_Block_Trikaal



Project Report

Complete documentation of our
problem, solution & development.



<https://drive.google.com/drive/folders/1hTTHVIDjegOB918AqDYhyQoiom11lQdL?usp=sharing>



Prototype Demonstration

RAKSHA-BLOCK working prototype
walkthrough



https://youtu.be/GBNe_w2-eEc



All data synthetic and clearly labelled · Model metrics in
models/model_metrics.json · Solver timings measured
on CPU on 23/09/2026 · Gemini is used for the
assistant chat only



raksha-block-trikaal.vercel.app
Scan to try the live prototype

<https://raksha-block-trikaal.vercel.app/>

Thank you: RAKSHA-BLOCK

AI recommends. The Section Controller decides.

Team TRIKAAL · SIH26027 · Ministry of Railways · Software